5.4.1 Writing expressive Given steps

Let's start with the `Given` step. This step should aim at getting the application in the appropriate state as quickly as possible. In this regard, it's fine to take a few shortcuts. For example, if you're testing a web application, you should try to imitate the user experience and actions, and so do as much as possible via the web interface. But if you need to set up test data in the database, it's fine to bypass the web interface and use a backend service to update the database.

The `Given` step, like the others, should reflect the business intent, or *what*, not the technical implementation, or *how*. For example, consider the following:

```
Given that an admin account is set up in the database
And I am logged in as admin
...
```

❷ You're only interested in the business intent expressed here.

❶ This sort of technical detail should not appear in the scenario.

In line ❶, too much detail is exposed about how the test is set up. This should be done silently, behind the scenes, both to avoid cluttering the scenario with unnecessary detail and to allow you to set it up as you see fit. All you're interested in at this level is the business context, which in this case is that the user is logged in with the administrator role ❷. The following version focuses the attention on the precondition in business terms:

```
Given I am logged in as an administrator
...
```

The `Given` step should also contain all the preconditions or steps that must have occurred *before* the action you're testing: no more, no less. For example, the following scenario is a little unclear:

```
Given that Bill registers for online banking
And that Bill opens the following accounts:
account | type    | balance
123456  | savings | 1000
123457  | current | 100
When Bill logs in
And Bill goes to the home page
And Bill views his accounts
Then Bill should see a list of his accounts:
account | type    | balance
123456  | savings | 1000
123457  | current | 100
```

The preconditions

The action under test (?)

The expected outcome

In this scenario, it's not clear what you're testing. Are you checking that Bill can log in successfully, or are you more interested in the accounts that he can view? In the latter case, you could rewrite this scenario as follows:

```
Given that Bill is registered for online banking
And that Bill has opened the following accounts:
account | type    | balance
123456  | savings | 1000
123457  | current | 100
```

❶ Putting these in the past tense makes it clear that they're preconditions.